Block practice · Bones, muscles and joints

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Identify bones and bone tissues on a diagram

Name joint types and compare their movement

State the function of a skeletal or joint part and explain effects of damage

Explain how biceps, triceps and tendons move the arm

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Practice workbook · Identify bones and bone tissues on a diagram

Naming bones of the skeleton and the tissues of a long bone from drawings or images.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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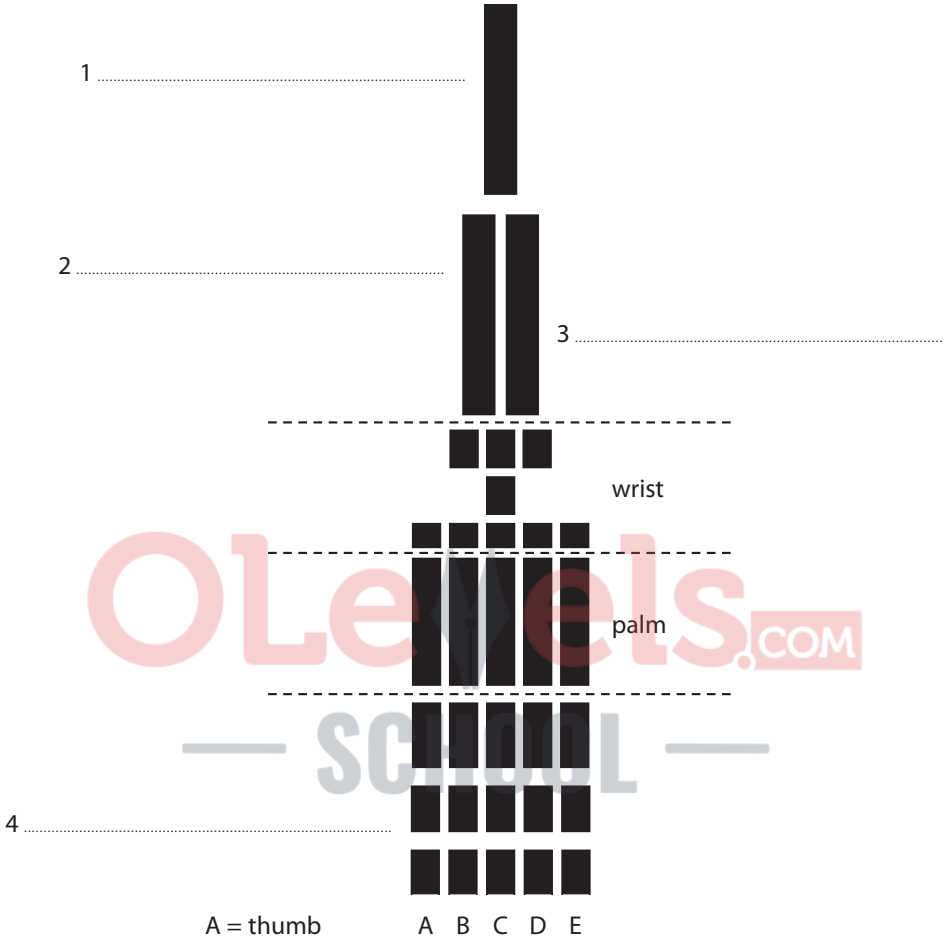
Type ID: HBI-SKELETON-BONE-IDENTIFY

1. Question 2 · part a.i · 4 marks

June 2021 Paper 1

Worked-example retry · Attempt part a.i only.

2 The diagram shows a model representing the arrangement of bones in the arm.



(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

(4)

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

(1)

(iii) Name two other types of synovial joint found in the skeleton.

(2)

1 .....

2 .....

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1. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

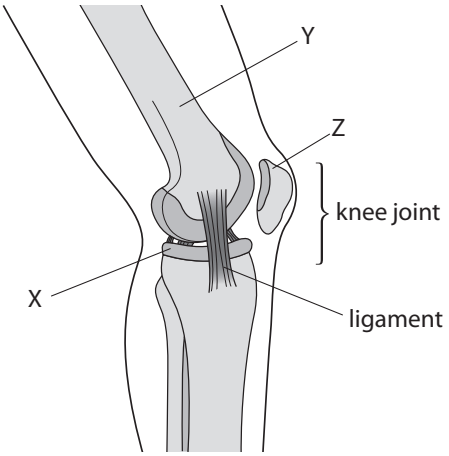
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2. Question 1 · part a.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part a.i only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

.....

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2. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

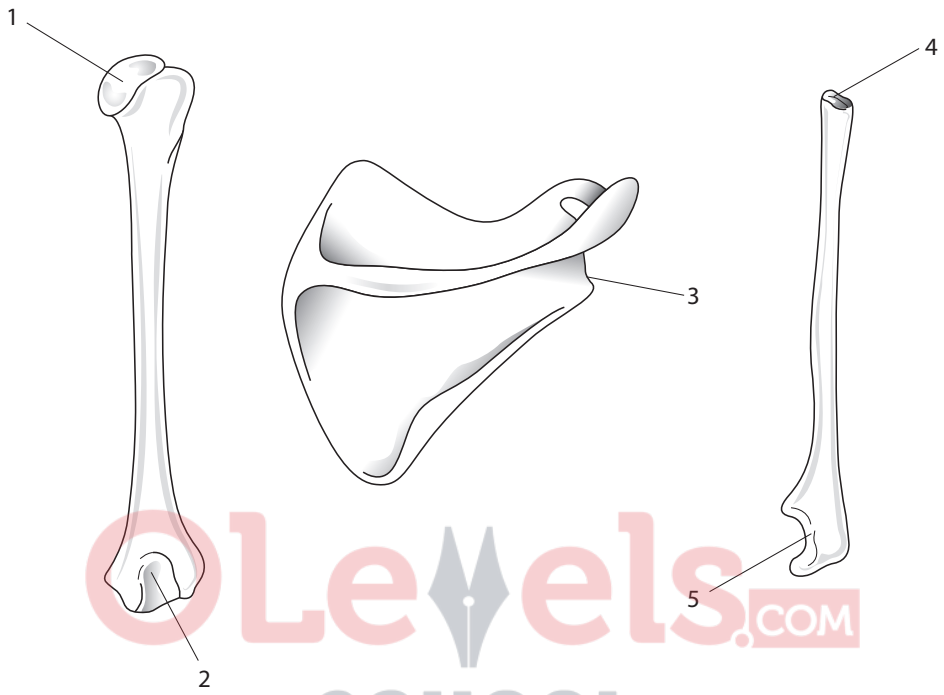
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3. Question 2 · part b.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.i only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

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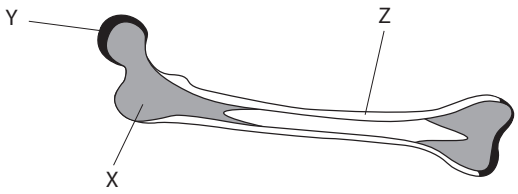
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3. Question 2 · part b.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.i only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

.....

(iv) Describe the function of the part labelled Y.

(2)

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3. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

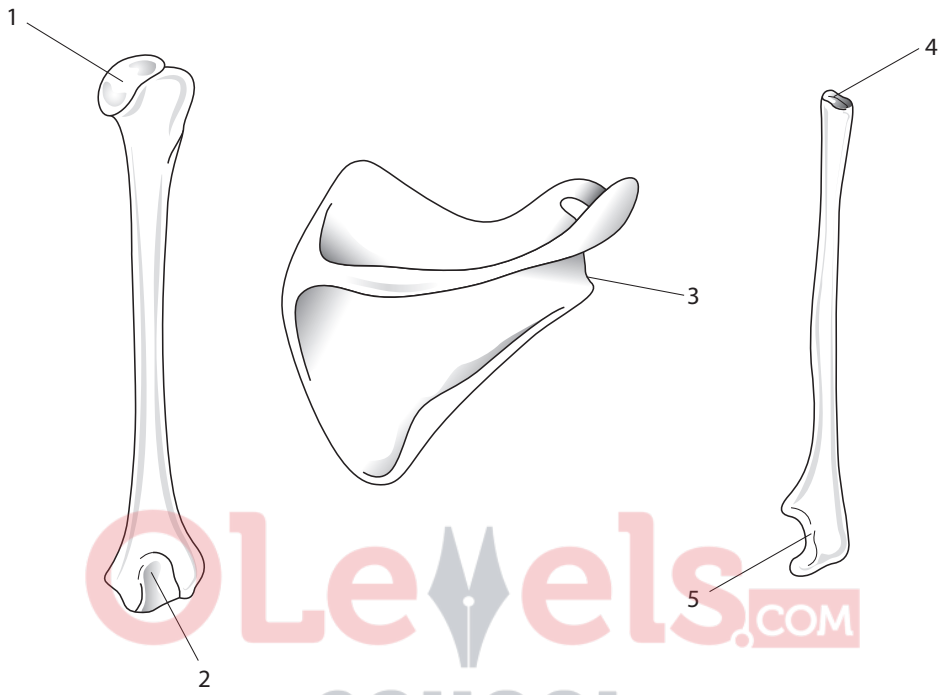
OLevels.com SCHOOL

4. Question 2 · part b.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- **A** 1 and 3
- **B** 1 and 5
- **C** 2 and 3
- **D** 2 and 4

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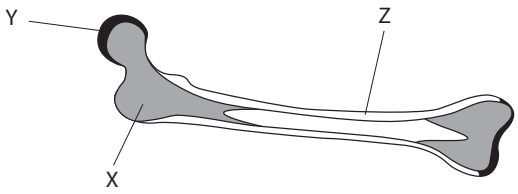
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4. Question 2 · part b.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

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(iv) Describe the function of the part labelled Y.

(2)

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4. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

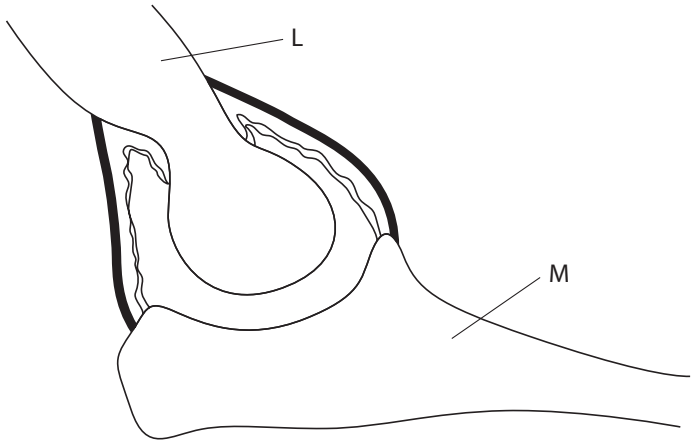
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5. Question 6 · part b.i · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

.....

.....

.....

.....

(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

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5. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

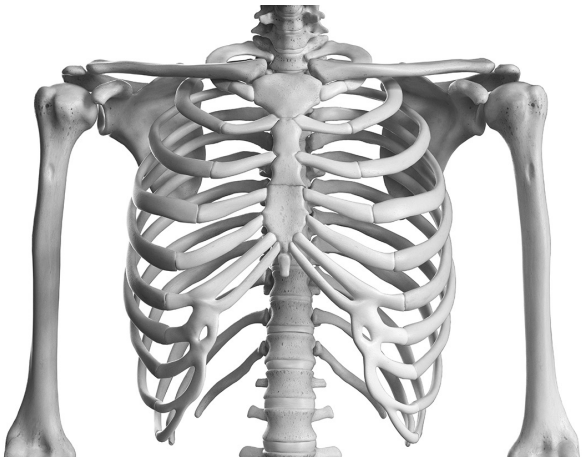
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6. Question 1 · part a.i · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

1 (a) The image shows part of the skeleton.



(Source: © SciePro/Shutterstock)

(i) Label the clavicle, scapula and vertebral column on the image. **(3)**

(ii) The vertebral column is part of the axial skeleton.

Describe the functions of the axial skeleton. **(2)**

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6. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 2 · part b.i · 2 marks

June 2023 Paper 1R

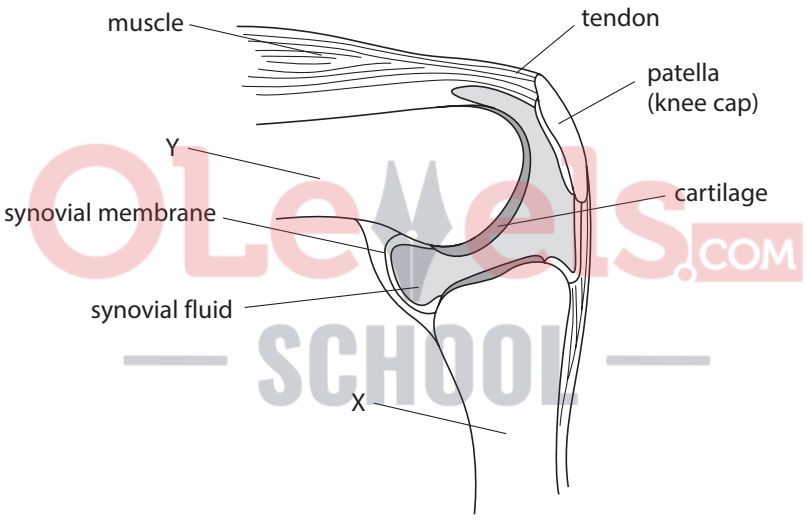
Further practice · Attempt part b.i only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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7. Question 2 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

(ii) Give the functions of tendons and cartilage.

(4)

tendons

cartilage

(iii) Give two structures, not shown on the diagram, that would be necessary for the joint to fully function.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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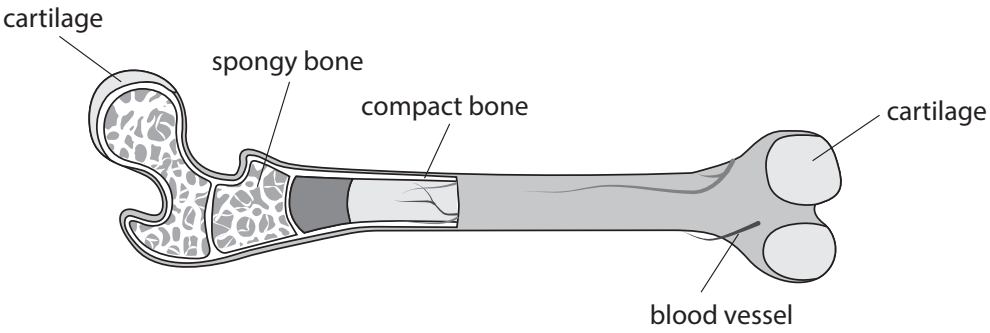
Working and answer

8. Question 1 · part a.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.i only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

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8. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Name joint types and compare their movement

What a joint is, hinge, ball-and-socket and gliding joints, where they occur and the movement each allows.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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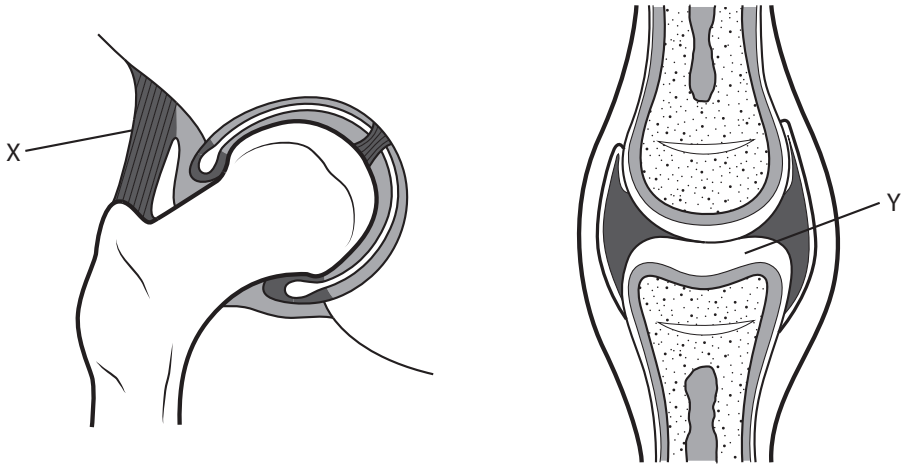
Type ID: HBI-JOINT-TYPE-MOVEMENT

1. Question 5 · part a.ii · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B. **(2)**

joint A .....

joint B .....

(ii) Describe the range of movement of joint A compared with the range of movement of joint B. **(2)**

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1. Question 5 · part a.ii · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

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(b) Part X and part Y play an important role in the function of the joint.

(i) Give the role of part X.

(1)

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.....

(ii) Give the role of part Y.

(1)

.....

.....

(Total for Question 5 = 6 marks)

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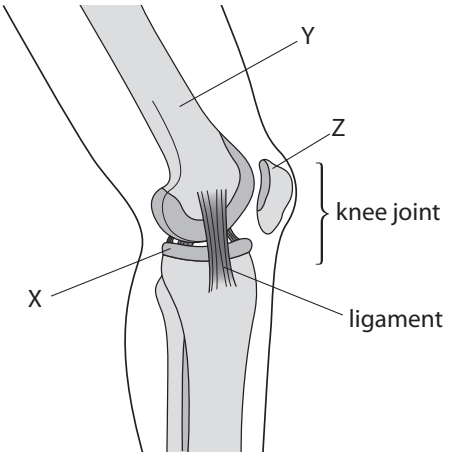
Working and answer

2. Question 1 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

.....

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2. Question 1 · part a.iii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.iii only.

(b) Aerobic respiration releases energy for muscles to contract.

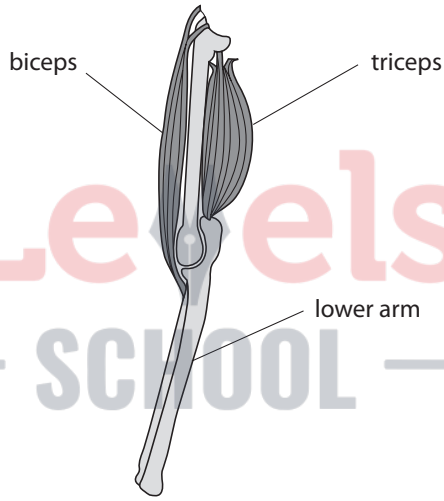
Complete the word equation for aerobic respiration.

(2)

glucose + ... → ..... + water

(c) Diagram 1 shows a model of the human arm, with the lower arm extended.

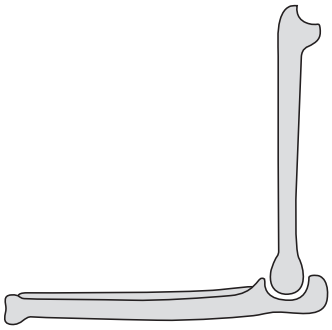
Diagram 1



Complete diagram 2 to show how the muscles change when the lower arm is raised.

(2)

Diagram 2



(Total for Question 1 = 9 marks)

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2. Question 1 · part a.iii · 1 mark · Working and answer

Working and answer

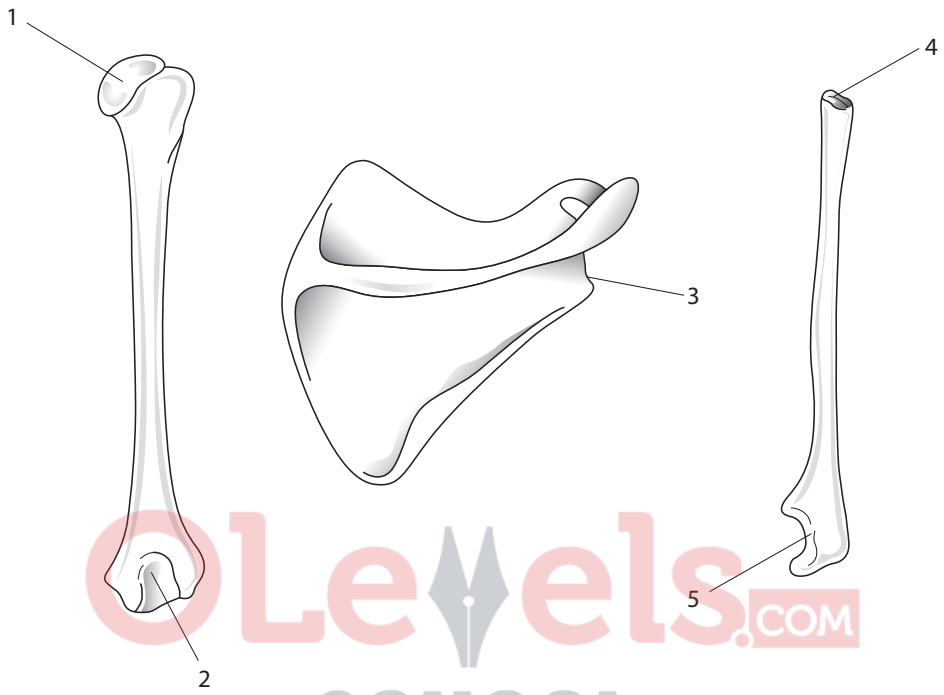
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3. Question 2 · part a · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

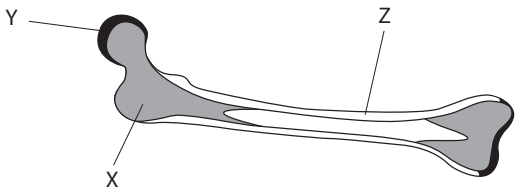
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3. Question 2 · part a · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

.....

(iv) Describe the function of the part labelled Y.

(2)

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3. Question 2 · part a · 1 mark · Working and answer

Working and answer

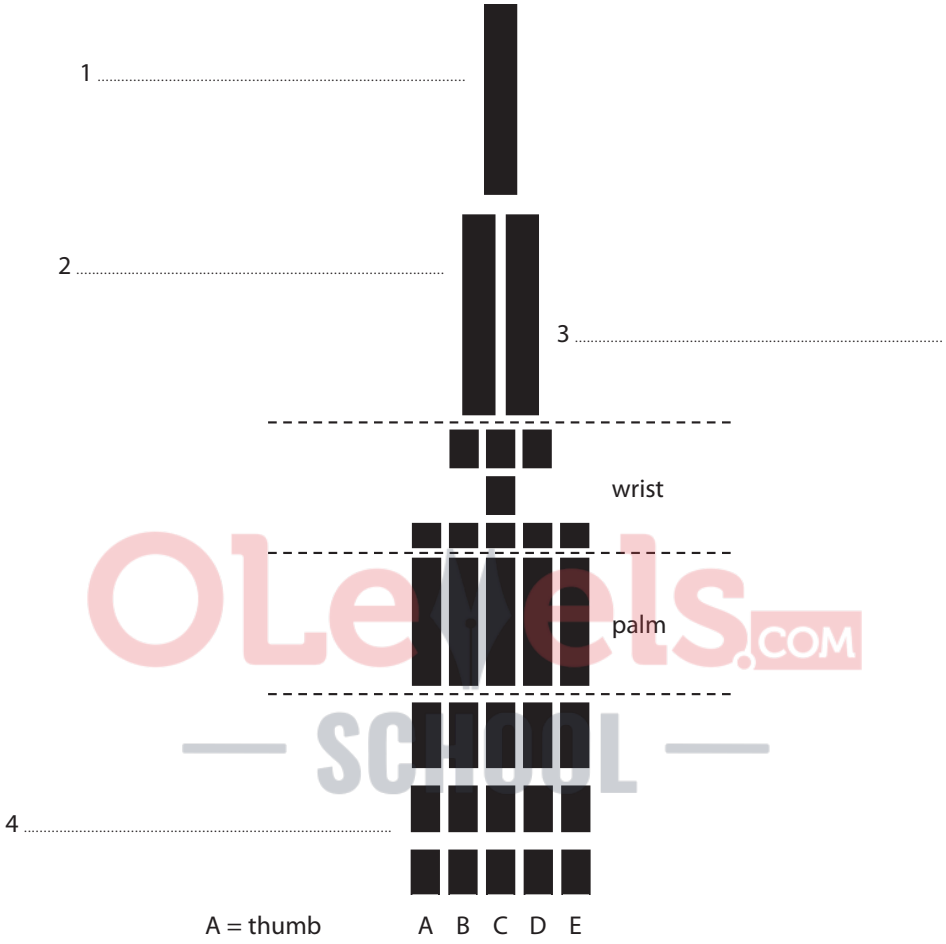
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4. Question 2 · part a.ii · 1 mark

June 2021 Paper 1

Further practice · Attempt part a.ii only.

2 The diagram shows a model representing the arrangement of bones in the arm.



(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

(4)

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

(1)

(iii) Name two other types of synovial joint found in the skeleton.

(2)

1 .....

2 .....

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4. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

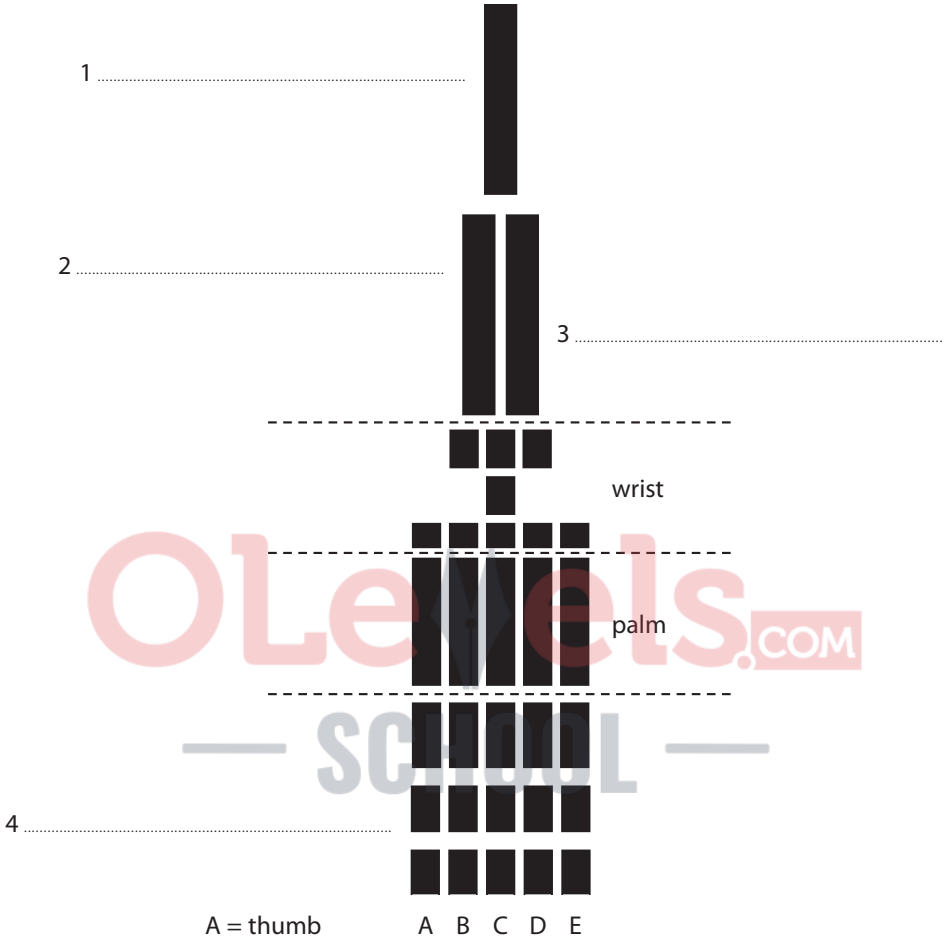
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5. Question 2 · part a.iii · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

2 The diagram shows a model representing the arrangement of bones in the arm.



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(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

(4)

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

(1)

(iii) Name two other types of synovial joint found in the skeleton.

(2)

1 .....

2 .....

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5. Question 2 · part a.iii · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

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(b) Vitamin D is important for the growth and development of healthy bones.

Explain how vitamin D helps the growth and development of bones.

(3)

(Total for Question 2 = 10 marks)

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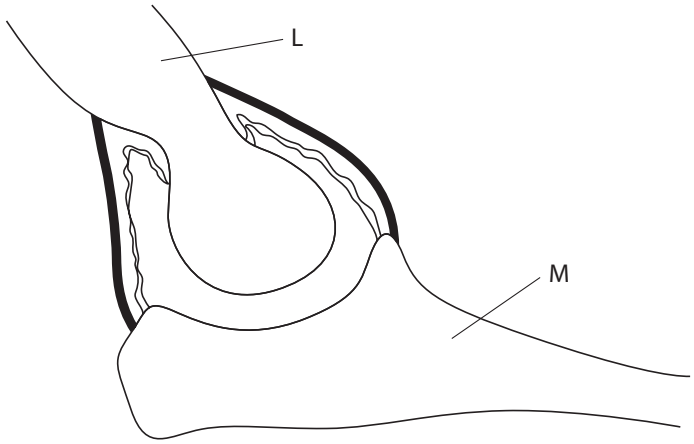
Working and answer

6. Question 6 · part a · 2 marks

November 2021 Paper 2

Further practice · Attempt part a only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

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.....

(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

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6. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 2 · part a · 2 marks

June 2023 Paper 1R

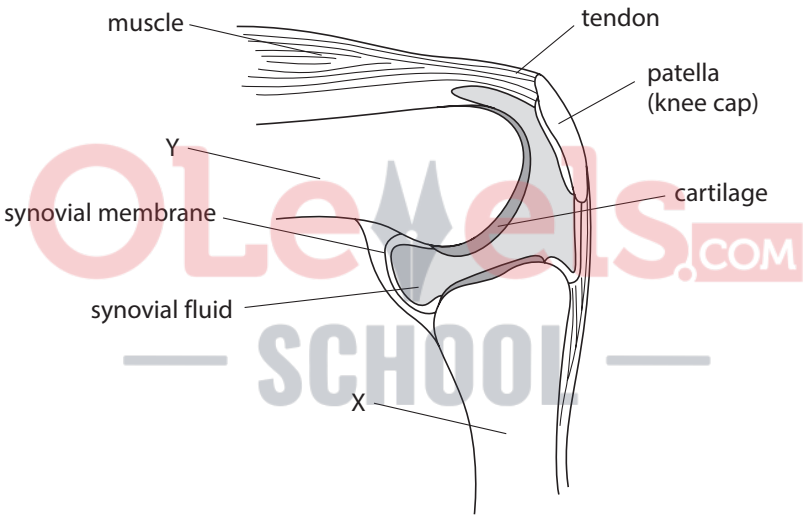
Further practice · Attempt part a only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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7. Question 2 · part a · 2 marks · Working and answer

Working and answer

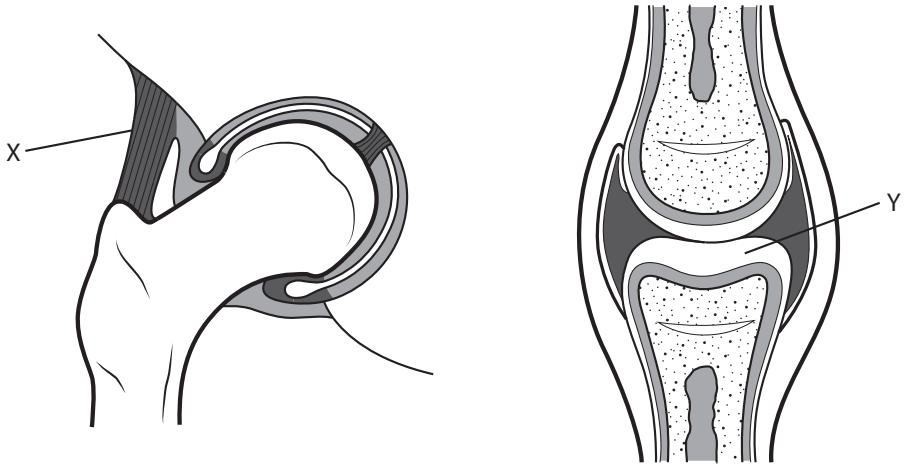
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8. Question 5 · part a.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.i only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B.

(2)

joint A .....

joint B .....

(ii) Describe the range of movement of joint A compared with the range of movement of joint B.

(2)

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8. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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Practice workbook · State the function of a skeletal or joint part and explain effects of damage

Roles of ligaments, tendons, cartilage, synovial fluid, bone tissues and the axial skeleton, and what happens when cartilage or collagen is lost.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SKELETAL-PART-FUNCTION

1. Question 2 · part b.ii · 4 marks

June 2023 Paper 1R

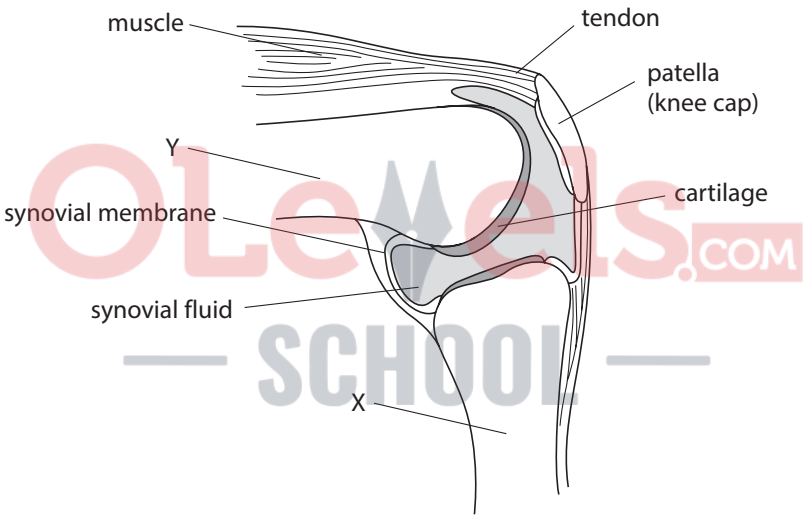
Worked-example retry · Attempt part b.ii only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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1. Question 2 · part b.ii · 4 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

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(ii) Give the functions of tendons and cartilage.

(4)

tendons

.....

.....

cartilage

.....

.....

(iii) Give two structures, not shown on the diagram, that would be necessary for the joint to fully function.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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1. Question 2 · part b.ii · 4 marks · Working and answer

Working and answer

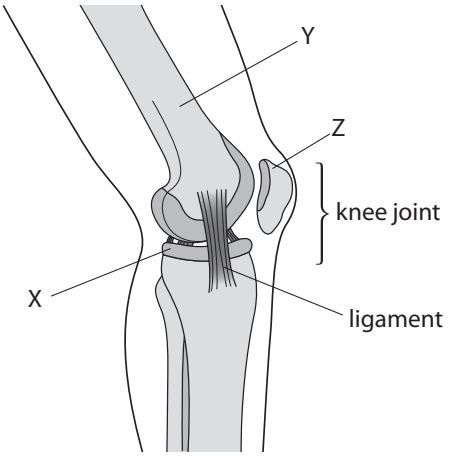
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2. Question 1 · part a.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

.....

2

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2. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

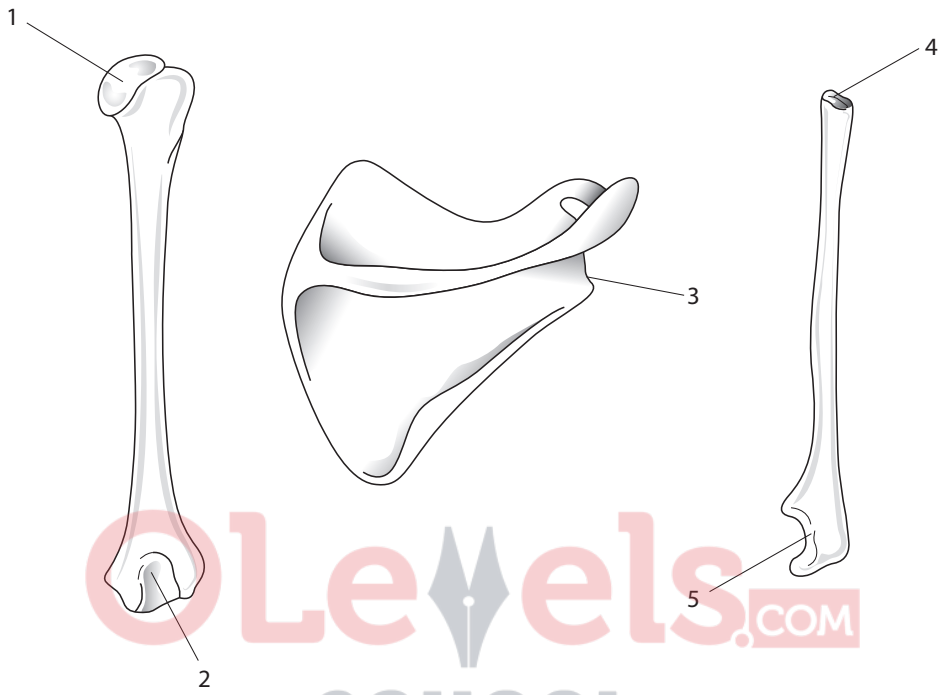
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3. Question 2 · part b.iv · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.iv only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

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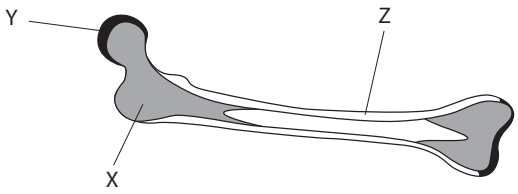
OLevels.com SCHOOL

3. Question 2 · part b.iv · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.iv only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

.....

(iv) Describe the function of the part labelled Y.

(2)

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3. Question 2 · part b.iv · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.iv only.

(c) Describe why a balanced diet is needed for the healthy development of bone and muscle.

(6)

(Total for Question 2 = 12 marks)

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3. Question 2 · part b.iv · 2 marks · Working and answer

Working and answer

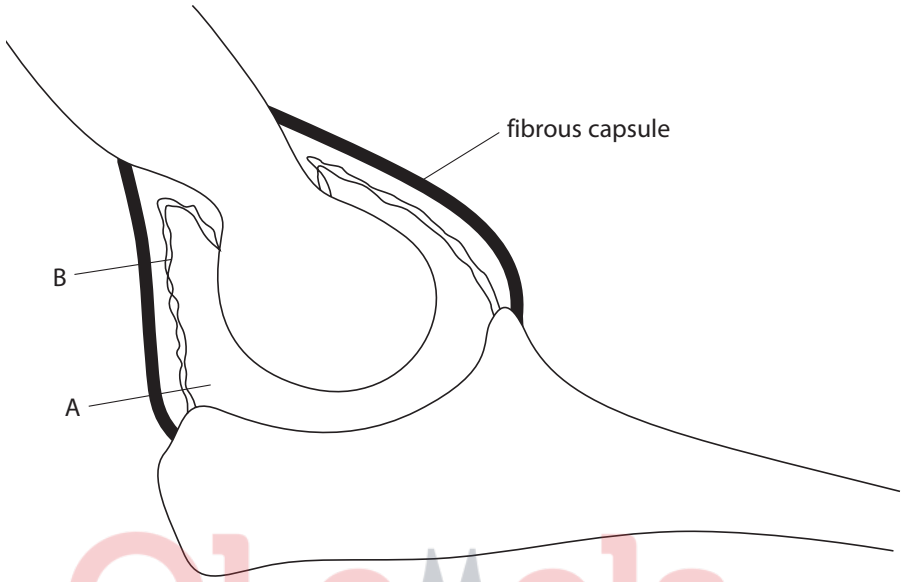
OLevels.com SCHOOL

4. Question 3 · part a.ii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.ii only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

.....

.....

.....

.....

(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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P 6 7 0 5 9 A 0 8 2 4

4. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

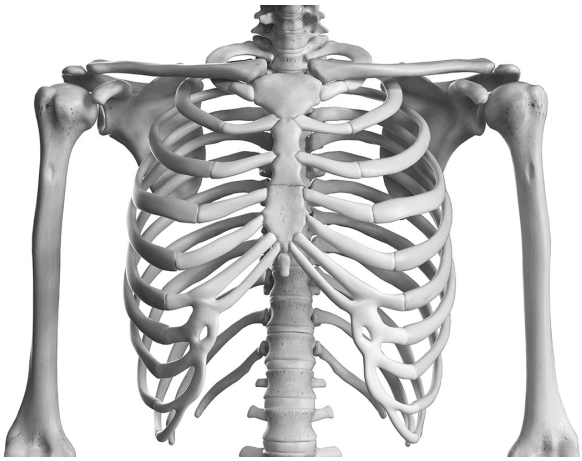
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5. Question 1 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

1 (a) The image shows part of the skeleton.



(Source: © SciePro/Shutterstock)

(i) Label the clavicle, scapula and vertebral column on the image.

(3)

(ii) The vertebral column is part of the axial skeleton.

Describe the functions of the axial skeleton.

(2)

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5. Question 1 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(b) Osteoporosis can affect the vertebral column.

Describe the symptoms of osteoporosis.

(3)

(Total for Question 1 = 8 marks)

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Working and answer

6. Question 2 · part b.iii · 2 marks

June 2023 Paper 1R

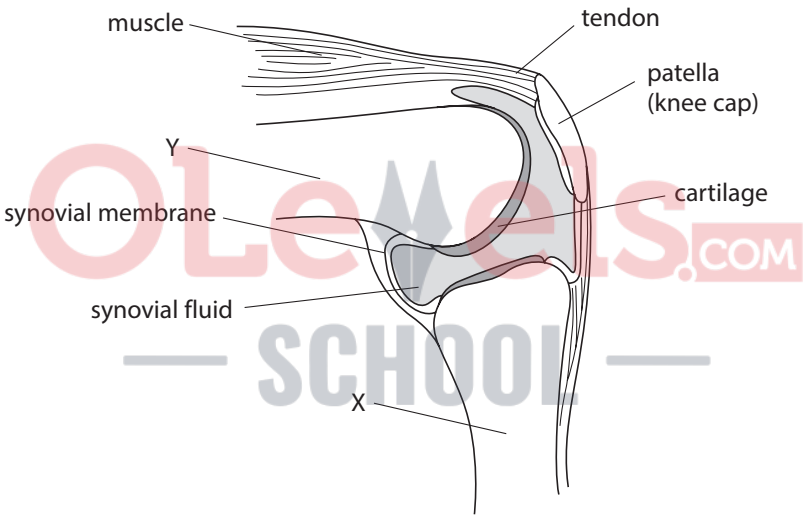
Further practice · Attempt part b.iii only.

2 The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

(2)

(b) The diagram shows a knee joint.



(i) Identify bone X and bone Y.

(2)

X .....

Y .....

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6. Question 2 · part b.iii · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.iii only.

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(ii) Give the functions of tendons and cartilage. **(4)**

tendons

cartilage

(iii) Give two structures, not shown on the diagram, that would be necessary for the joint to fully function. **(2)**

1 .....

2 .....

(Total for Question 2 = 10 marks)

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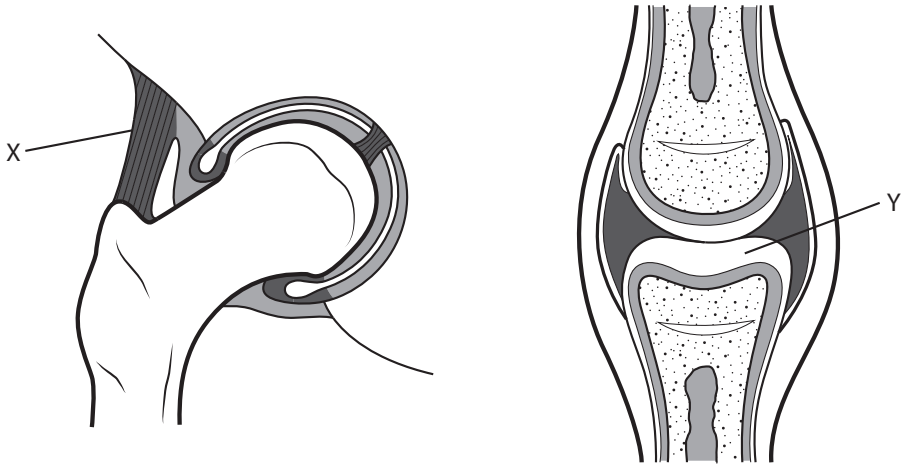
Working and answer

7. Question 5 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B. **(2)**

joint A

joint B

(ii) Describe the range of movement of joint A compared with the range of movement of joint B. **(2)**

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P 7 4 4 9 2 A 0 1 4 2 4

7. Question 5 · part b.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.i only.

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(b) Part X and part Y play an important role in the function of the joint.

(i) Give the role of part X.

(1)

.....

.....

(ii) Give the role of part Y.

(1)

.....

.....

(Total for Question 5 = 6 marks)

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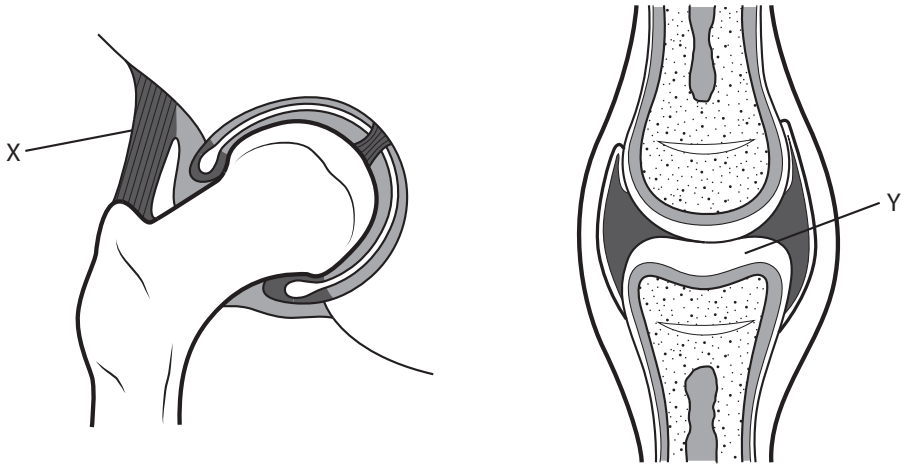
Working and answer

8. Question 5 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

5 The diagram shows two different types of synovial joint in the human body.



Joint A Joint B

(a) (i) Name the type of joint shown by joint A and by joint B. **(2)**

joint A .....

joint B .....

(ii) Describe the range of movement of joint A compared with the range of movement of joint B. **(2)**

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8. Question 5 · part b.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(b) Part X and part Y play an important role in the function of the joint.

(i) Give the role of part X.

(1)

.....

.....

(ii) Give the role of part Y.

(1)

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(Total for Question 5 = 6 marks)

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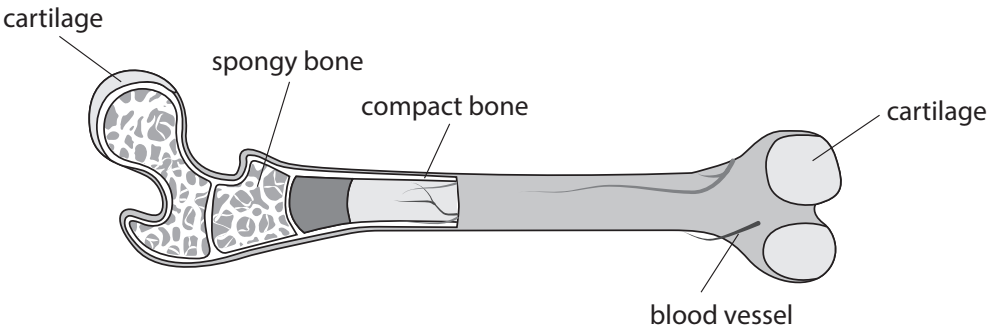
Working and answer

9. Question 1 · part a.ii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

.....

.....

.....

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9. Question 1 · part a.ii · 3 marks · Working and answer

Working and answer

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10. Question 8 · part a.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

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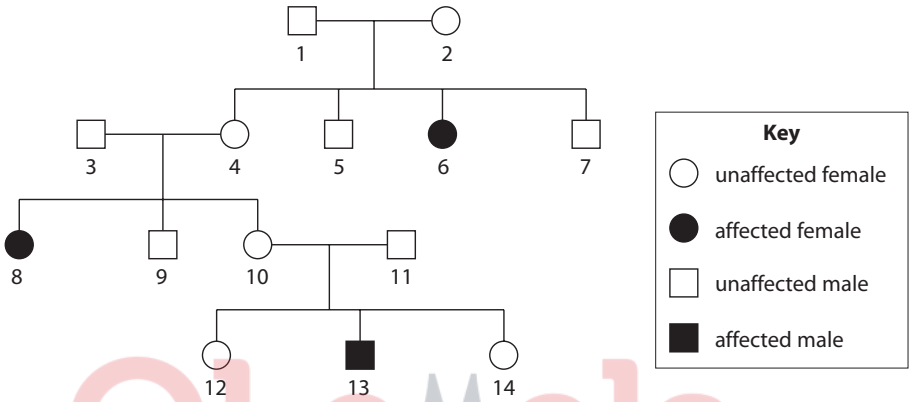
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10. Question 8 · part a.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

(b) The diagram shows the pedigree of a family affected by alkaptonuria.



(i) Use the family pedigree to explain whether the alkaptonuria condition is sex-linked.

(4)

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10. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

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11. Question 4 · part e · 2 marks

June 2019 Paper 2

Further practice · Attempt part e only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

.....

.....

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11. Question 4 · part e · 2 marks

June 2019 Paper 2

Further practice · Attempt part e only.

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(d) Describe a method that Goldberger could use to show that the symptoms of his patients were caused by a lack of vitamin B in their diet.

(3)

(e) Scurvy is another deficiency disease.

Scurvy affects the production of a protein called collagen.

Collagen is a component of many body tissues including cartilage.

Explain why a person with scurvy often experiences pain when walking.

(2)

(Total for Question 4 = 9 marks)

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11. Question 4 · part e · 2 marks · Working and answer

Working and answer

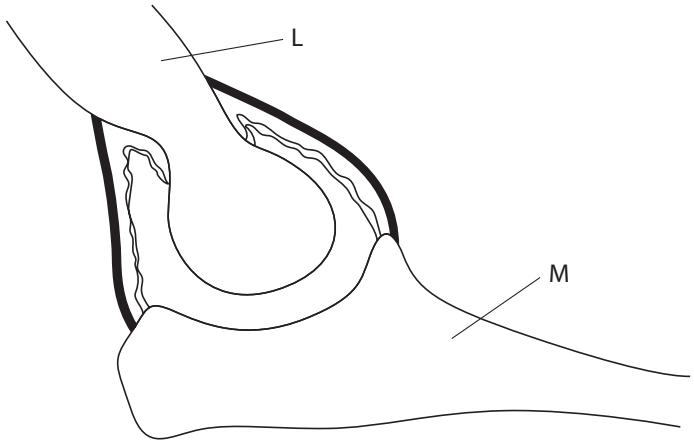
OLevels.com SCHOOL

12. Question 6 · part c · 2 marks

November 2021 Paper 2

Further practice · Attempt part c only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

.....

.....

.....

.....

(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

20

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12. Question 6 · part c · 2 marks

November 2021 Paper 2

Further practice · Attempt part c only.

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(c) Explain the effects on a person if the cartilage in this joint is missing or damaged.

(2)

(d) Tendons are important in the functioning of a joint.

Explain the importance of tendons in the movement of bone M.

(4)

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(Total for Question 6 = 12 marks)

12. Question 6 · part c · 2 marks · Working and answer

Working and answer

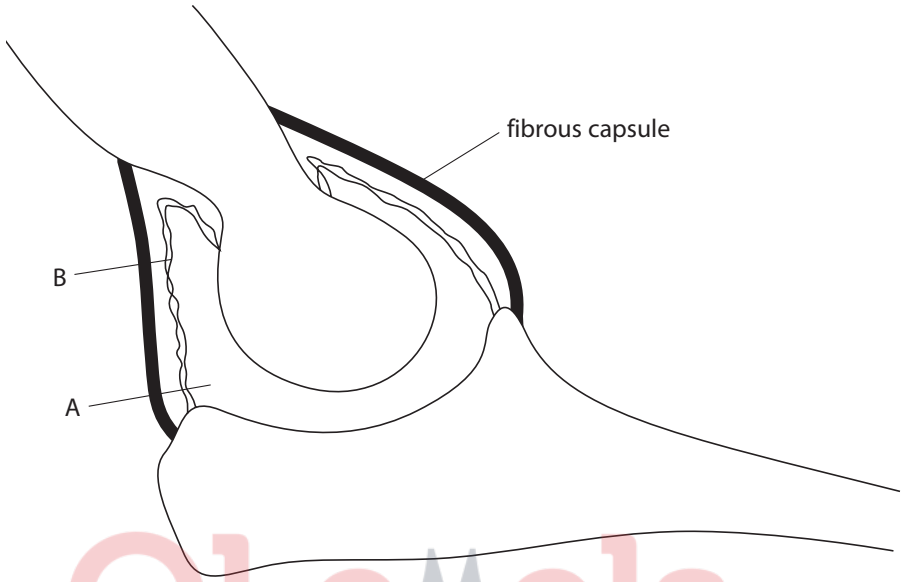
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13. Question 3 · part b.ii · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

.....

.....

.....

.....

(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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13. Question 3 · part b.ii · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Cartilage often breaks down as people become older.

Explain the effects this has on the working of the joint.

(2)

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(c) Explain the role of tendons in the movement of the lower arm.

(5)

(Total for Question 3 = 13 marks)

13. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how biceps, triceps and tendons move the arm

Antagonistic muscle action and tendons moving the forearm at the elbow.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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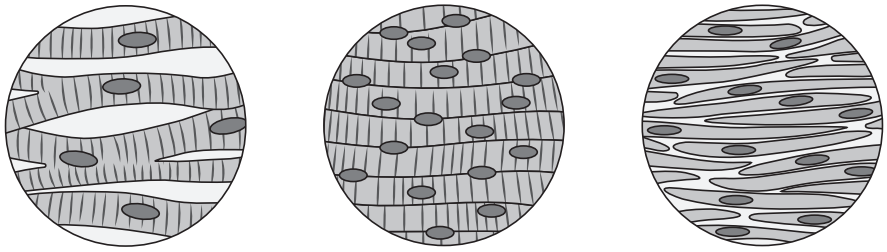
Type ID: HBI-ANTAGONISTIC-MUSCLE-MOVEMENT

1. Question 6 · part c · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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1. Question 6 · part c · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

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(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

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1. Question 6 · part c · 4 marks

November 2024 Paper 1

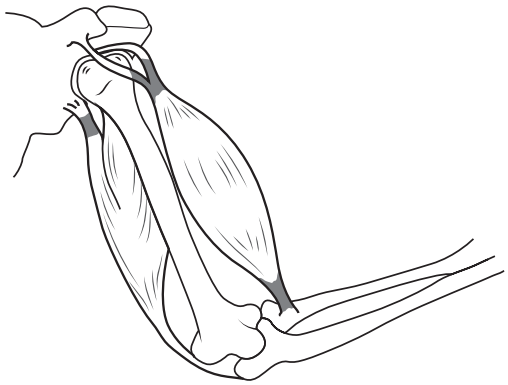
Worked-example retry · Attempt part c only.

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(c) The diagram shows two muscles in the arm.



Explain how these muscles help to raise the forearm.

(4)

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(Total for Question 6 = 17 marks)

1. Question 6 · part c · 4 marks · Working and answer

Working and answer

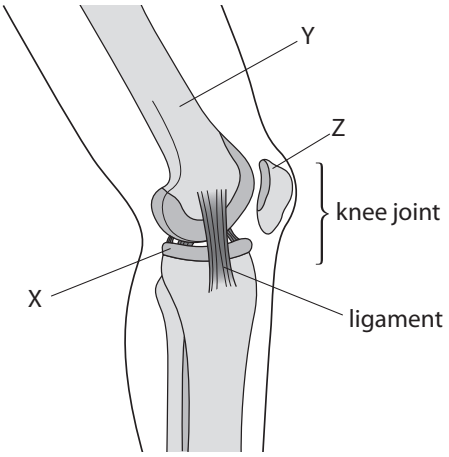
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2. Question 1 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

.....

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2. Question 1 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

(b) Aerobic respiration releases energy for muscles to contract.

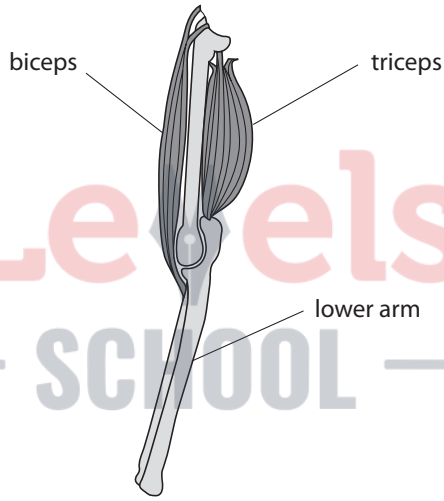
Complete the word equation for aerobic respiration.

(2)

glucose + ... → ..... + water

(c) Diagram 1 shows a model of the human arm, with the lower arm extended.

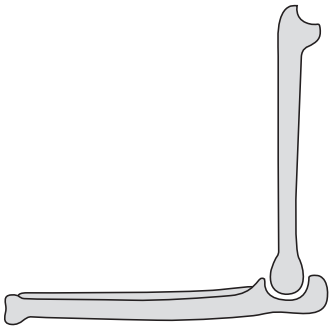
Diagram 1



Complete diagram 2 to show how the muscles change when the lower arm is raised.

(2)

Diagram 2



(Total for Question 1 = 9 marks)

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2. Question 1 · part c · 2 marks · Working and answer

Working and answer

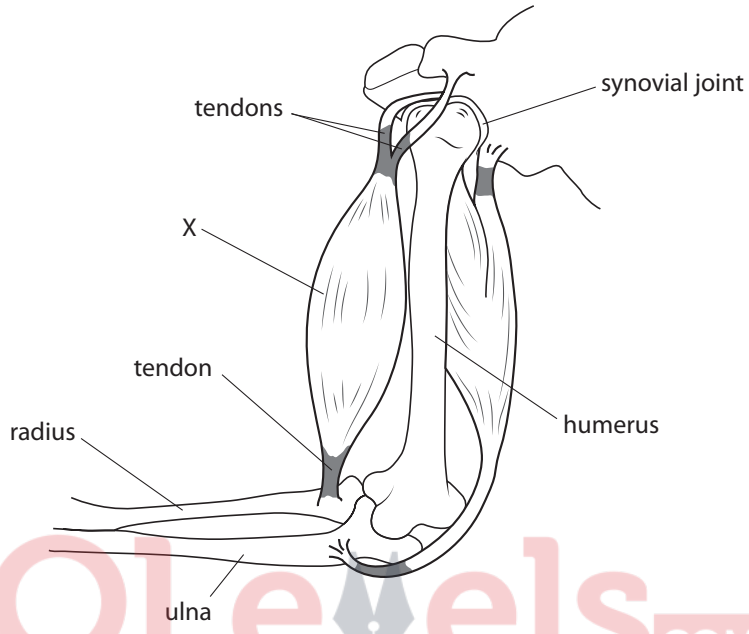
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3. Question 7 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

7 The diagram shows the bones of a human arm and the muscles that move the lower arm.



(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

(1)

.....

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

(2)

.....

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3. Question 7 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

(iii) Explain how the contraction of muscle X causes the lower arm to be raised.

(3)

.....

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.....

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.....

(b) Draw a labelled diagram to show the internal structure of the synovial joint.

(6)

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(Total for Question 7 = 12 marks)

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3. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

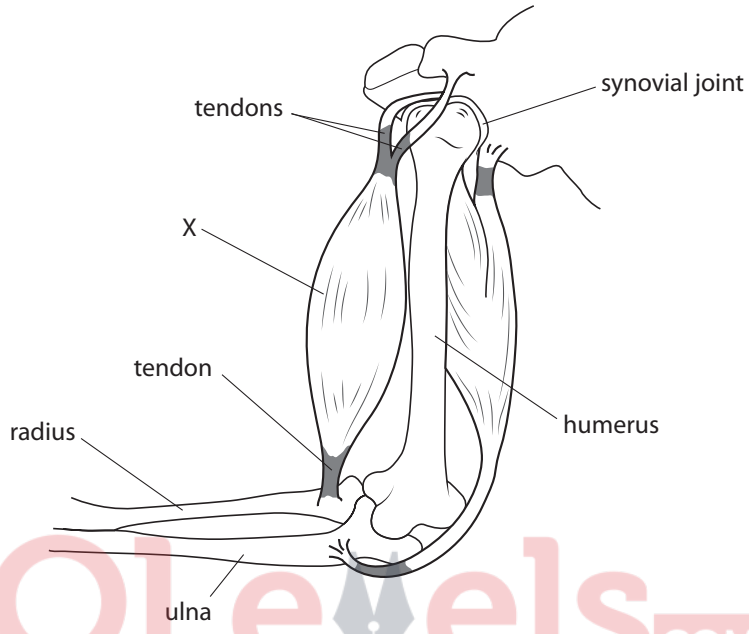
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4. Question 7 · part a.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

7 The diagram shows the bones of a human arm and the muscles that move the lower arm.



(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

(1)

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

(2)

20

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4. Question 7 · part a.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

(iii) Explain how the contraction of muscle X causes the lower arm to be raised.

(3)

(b) Draw a labelled diagram to show the internal structure of the synovial joint.

(6)

(Total for Question 7 = 12 marks)

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4. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

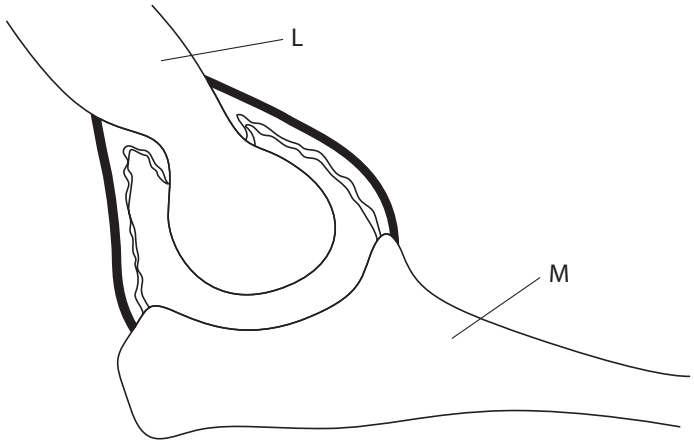
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5. Question 6 · part d · 4 marks

November 2021 Paper 2

Further practice · Attempt part d only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

.....

.....

.....

.....

(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

20

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5. Question 6 · part d · 4 marks

November 2021 Paper 2

Further practice · Attempt part d only.

(c) Explain the effects on a person if the cartilage in this joint is missing or damaged.

(2)

(d) Tendons are important in the functioning of a joint.

Explain the importance of tendons in the movement of bone M.

(4)

(Total for Question 6 = 12 marks)

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5. Question 6 · part d · 4 marks · Working and answer

Working and answer

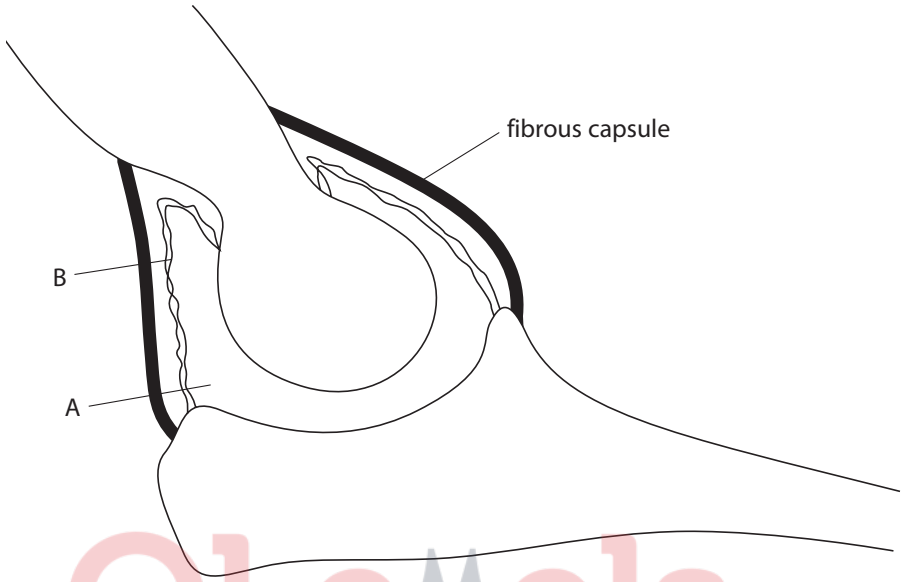
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6. Question 3 · part c · 5 marks

January 2022 Paper 1

Further practice · Attempt part c only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

.....

.....

.....

.....

(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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6. Question 3 · part c · 5 marks

January 2022 Paper 1

Further practice · Attempt part c only.

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(ii) Cartilage often breaks down as people become older.

Explain the effects this has on the working of the joint.

(2)

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(c) Explain the role of tendons in the movement of the lower arm.

(5)

(Total for Question 3 = 13 marks)

6. Question 3 · part c · 5 marks · Working and answer

Working and answer

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7. Question 10 · part b · 3 marks

June 2022 Paper 1R

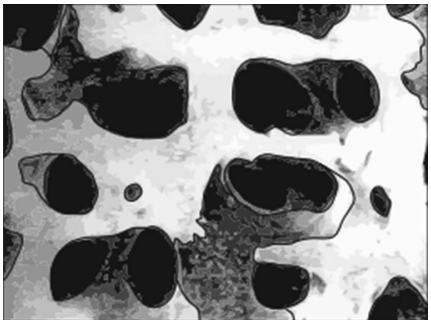
Further practice · Attempt part b only.

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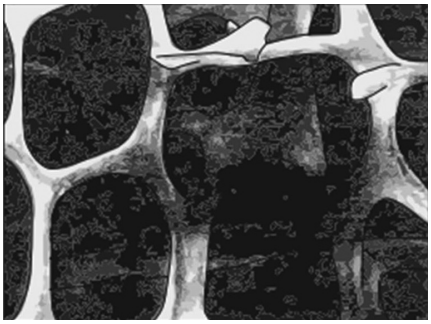
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10 (a) The photographs show the structure of healthy bone and bone that is damaged by osteoporosis.



healthy bone



bone damaged by osteoporosis

Describe how osteoporosis might affect an individual.

(3)

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(b) Explain how voluntary muscles and bones bring about movement in the elbow joint.

(3)

(Total for Question 10 = 6 marks)

7. Question 10 · part b · 3 marks · Working and answer

Working and answer

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8. Question 1 · part whole · 6 marks

June 2025 Paper 1

Further practice · Attempt part whole only.

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1 The sentences below are about muscles and movement of the arm.

Complete the sentences by using words from the box.

(6)

antagonistic	ball and socket	biceps	cartilage
complementary	fixed	hinge	ligament
tendons	triceps		

To cause the movement of a joint, muscles work in pairs, known as

..... pairs of muscles.

The elbow is a type of synovial joint known as a ..... joint.

This joint allows the forearm to be raised when the .....

muscle contracts, and to be straightened when the ..... muscle contracts.

Both of these muscles are connected to bones by ..... .

Smooth movement of the joint is possible because of the presence of

..... covering the ends of the bones.

(Total for Question 1 = 6 marks)

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8. Question 1 · part whole · 6 marks · Working and answer

Working and answer

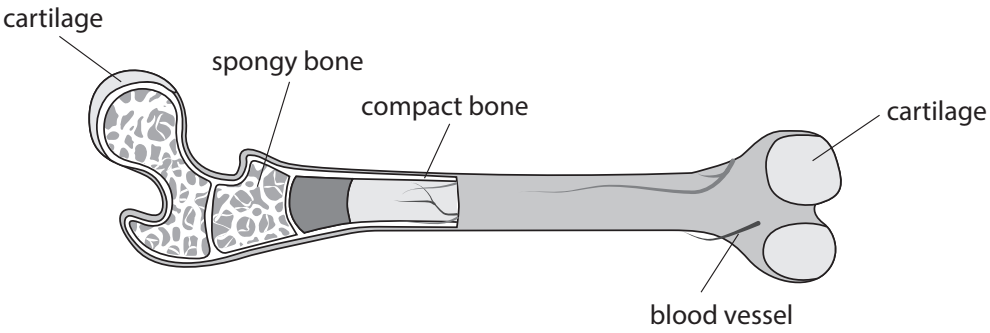
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9. Question 1 · part b.i · 4 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

.....

.....

.....

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9. Question 1 · part b.i · 4 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

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(b) Bones are moved by muscles.

(i) Use the words in the box to complete the passage about bones and muscles.

(4)

cartilage	contracts	ligaments	lowers	pulls
pushes	raises	relaxes	tendons	

Muscles are attached to bones by ..... .

To raise the lower arm, the biceps ..... and the triceps ..... .

The biceps ..... on the radius, which raises the lower arm.

(ii) Energy is transferred by the biceps muscle when raising the lower arm.

The biceps muscle applies a force of 125 N to raise the lower arm a distance of 0.50 m.

Use this formula to calculate the energy transferred by the biceps muscle when raising the lower arm 0.50 m.

energy transferred (J) = force (N) × distance (m)

(2)

energy transferred = ... J

(Total for Question 1 = 12 marks)

9. Question 1 · part b.i · 4 marks · Working and answer

Working and answer

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